Part A: Conceptual Understanding (Theory)

1. What are Supervised Learning Algorithms?

Supervised learning algorithms learn patterns from labelled data, where input features and the correct target are provided. They use these patterns to predict outputs for new data. In this project, house features are used to predict house prices.

2. Difference between Regression Algorithms and Classification Algorithms

Regression algorithms predict continuous numerical values, such as a house price. Classification algorithms predict categories or labels, such as low-price, medium-price, or high-price houses. Predicting an exact house price is a regression task.

3. Explain Simple Linear Regression

Simple Linear Regression describes the relationship between one independent variable and one dependent variable using a straight line. Its equation is $\hat{y} = \beta_0 + \beta_1 x$, where β_0 is the intercept and β_1 is the slope. Here, house area can be used to predict house price.

4. List and explain the assumptions of Linear Regression

The main assumptions are linearity, independence of errors, constant error variance (homoscedasticity), and approximately normal residuals for statistical inference. In multiple regression, predictors should also not have excessive multicollinearity. These assumptions can be checked using diagnostic plots and appropriate tests.

5. What is Bias–Variance Trade-Off?

Bias is error caused by a model being too simple, while variance is sensitivity to the particular training data. A high-bias model may underfit, while a high-variance model may overfit. The goal is to balance both so the model predicts unseen house prices accurately.

6. Explain Overfitting and Underfitting with examples

Overfitting occurs when a model learns noise in the training data and performs worse on unseen data; a high-degree polynomial may overfit house prices. Underfitting occurs when a model is too simple to capture the pattern and performs poorly on both training and test data. Comparing training and test metrics helps identify these problems.